Explaining Pain by Combining Deep Learning Models and Physiology-Driven Ensembles using PPG, EDA, and Respiration

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Abstract

Accurate pain assessment remains a major challenge in clinical practice, particularly in non-communicative patients where self-report is unfeasible. This study presents a hybrid, explainable approach for automatic pain classification from physiological signals, developed in the context of the AI4PAIN 2025 Challenge. We address key limitations of the dataset (limited size, signal discontinuity, and quality degradation) by combining deep learning (DL) for pattern discovery with classical machine learning (ML) for robust classification. In Stage 1, a CNN-LSTM model with attention and explainability modules, Gradient-weighted Class Activation Maps (GradCAM) and feature ablation, identifies temporally and modality relevant signal patterns. Guided by these insights, Stage 2 extracts a compact set of physiologically grounded features from photoplethysmography (PPG), electrodermal activity (EDA), and chest respiratory effort (RESP) signals. Stage 3 employs an ensemble of optimized ML models (CatBoost, XGBoost, LightGBM) trained on these features, achieving 0.84 F1-macro on validation data. Explainability analysis confirms that the model captures clinically relevant pain markers, such as sympathetic arousal (EDA), disrupted pulse variability, in terms of amplitude and width, and altered respiratory patterns. Despite a validation-to-test performance drop (0.14), our XAI-guided methodology demonstrates strong interpretability, highlighting its

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ICMI Companion '25, Canberra, ACT, Australia
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ACM ISBN 979-8-4007-2076-5/25/10
https://doi.org/10.1145/3747327.3764789

potential for trustworthy pain assessment in real-world clinical settings.

Keywords

Objective Pain Assessment; XAI; Feature Engineering; Ensemble Learning; PPG, EDA, RESP signal processing

ACM Reference Format:
Miguel Javierre del Río, Pablo Armañac Julián, Rodrigo Lozano Puñet, Diego Cajal Orleans, Nayan Wadhwani, Jesús Lázaro, and Raquel Bailón Luesma. 2025. Explaining Pain by Combining Deep Learning Models and Physiology-Driven Ensembles using PPG, EDA, and Respiration. In *Companion Proceedings of the 27th International Conference on Multimodal Interaction (ICMI Companion '25), October 13–17, 2025, Canberra, ACT, Australia.* ACM, New York, NY, USA, 10 pages. https://doi.org/10.1145/3747327.3764789

1 Introduction

Accurate pain assessment is a fundamental challenge in medicine that often guides diagnosis, treatment, and patient care. The current methods for pain evaluation rely on patient self-report using simple rating scales such as the Numeric Rating Scale (NRS) or Visual Analog Scale (VAS). While effective in many cases, self-report measures become impossible or unreliable in patients who cannot communicate (such as infants, mechanically ventilated patients, or individuals with cognitive impairments, among others) [30]. Moreover, even when patients can report, their ratings are subjective and can be influenced by bias, anxiety, or context [28]. These limitations motivate the search for objective, physiology-based pain biomarkers that do not depend on self-perception feedback.

One promising approach is to assess the autonomic nervous system (ANS), which reacts to noxious stimuli. Pain triggers a cascade of ANS responses, particularly a sympathetic "fight-or-flight" reaction [22]. This response manifests in measurable changes across multiple biomedical signals. For example, electrodermal activity (EDA) rises due to sweat gland activation (ANS innervated), providing a direct index of sympathetic arousal [22]. Pulse photoplethysmography (PPG) captures blood volume pulses, from which heart rate and its variability (HRV) can be derived; notably, HRV reflects the dynamic interplay between sympathetic and parasympathetic control of the heart [39], since the sinoatrial node of the heart is also innervated by the vagus nerve from the ANS. Respiration is also controlled by the parasympathetic activity, and it is likewise modulated during painful experiences: acute pain may provoke an inspiratory gasp or breath-hold, whereas persistent pain can lead to rapid, shallow breathing (hyperventilation) [19]. Given that each modality offers a different window onto the pain response, an integrated multimodal approach may provide a more robust and comprehensive assessment than any single-signal analysis.

Because of the above mentioned hypothesis, Fernandez Rojas et al. published a multimodal dataset composed of different physiological signals recorded from subjects during electrical pain [34]. After the success of the AI4PAIN Challenge in 2024 [33] the AI4PAIN 2025 Challenge [14] is built around this dataset, which includes synchronized recordings from EDA, PPG, chest respiratory effort (RESP), and peripheral blood oxygen saturation (SpO_2) signals. The original rules of the challenge limited the use of pre-trained models. This restriction combined with the small amount of data available remains a considerable limitation for developing a robust deep learning (DL) approach from scratch. Moreover, data is organized in short and discontinuous segments, which suffer from quality issues, such as saturation. To address these problems, we propose a hybrid and explainable framework organized into three complementary stages. First, we take advantage of a deep neural architecture to discover temporal patterns across the multi-modal data, guided by explainable AI techniques. Subsequently, a set of physiologically meaningful features was extracted from the regions identified as informative. Then, these features were used as input to a set of optimized classical machine learning models for robust pain classification. This approach combines the pattern discovery capabilities of deep learning with the explainability of feature-based models, promoting both understandability and generalization.

2 Materials and Methods

Our methodology was designed to be robust to the characteristics of this dataset, specifically to its segmented nature and signal quality issues. The overall pipeline is a three-stage process that uses explainable deep learning (DL-XAI) for pattern discovery, feature extraction from found patterns, and a set of classical prediction models based on extracted features.

2.1 Dataset

The dataset, provided by the AI4PAIN 2025 Challenge, contains segments of EDA, PPG, and RESP signals with a sampling rate of 100 Hz from 65 subjects, with a 41/12/12 train/validation/test split. Labels correspond to No-Pain (NP), Low-Pain (LP), and High-Pain (HP) self-reported and induced by transcutaneous electrical nerve stimulation (TENS) [34]. Two significant data quality problems were found in the recorded signals: repeated EDA signal saturation and artifactual PPG segments. These artifacts and signal discontinuity greatly constrain traditional DL end-to-end methods. SpO_2 was not included in our analysis due to the dynamics of peripheral oxygen saturation being much slower than the length of the segments of the protocol. Given that our analysis is based on 10-second segments, SpO_2 does not have the necessary temporal resolution to provide a meaningful variation within such short windows [11].

2.2 Proposed Hybrid Methodology

We propose a hybrid and explainable methodology that integrates deep learning (DL) with classical machine learning (ML). This approach, is structured as a three-stage pipeline. In the first stage, a deep neural network on filtered physiological data was not trained with the primary goal of achieving high classification performance, but to uncover relevant patterns within the data. The most informative time windows, signal modalities, and temporal dynamics associated with pain levels were selected by applying explainable AI techniques: both Gradient-weighted Class Activation Maps (Grad-CAM) and feature ablation. These findings guided the second stage of our pipeline, where a set of physiology-driven features was extracted and used to train a set of ensemble models (CatBoost, XGBoost, LightGBM). This sequential process allows us to combine the pattern discovery capacity of DL with the robustness and interpretability of classical ML models, resulting in a generalizable and explainable framework for pain recognition.

2.2.1 Stage 1: Pattern Discovery in Physiological Signals via DL and XAI. In this stage, a deep neural network was designed and trained for pain classification from physiological signals. This approach has been explored in the literature using EDA, PPG, and respiratory signals for automated pain recognition [38, 41]. This work is particularly focused on finding meaningful patterns rather than purely optimizing classification accuracy. The architecture combines convolutional and recurrent components with an attention mechanism, as this hybrid design has shown effectiveness in modeling temporal dependencies in physiological signals [13, 21].

The network begins with a stack of three 1D convolutional layers with batch normalization and dropout, applied to each input segment of synchronized PPG, EDA, and respiratory signals. The extracted features are then passed through a bidirectional Long Short Term (LSTM) layer, which captures temporal dependencies across modalities. An attention mechanism computes soft temporal weights, allowing the model to focus on the most relevant time steps for classification. The final context vector is passed through a feed-forward classifier to predict one of three pain levels.

The model is trained using the Adam optimizer with an initial learning rate of 0.01 and cosine annealing scheduler over 200 epochs. Class imbalance is addressed through weighted crossentropy loss computed from inverse class frequencies in the training set, a strategy known to improve performance under skewed class distributions in deep networks [7]. Training employs early stopping based on validation loss with a patience of 10 epochs to prevent overfitting, and segmentation into overlapping 10 second windows with 25% of overlap (1000 samples at 250 Hz sampling rate).

To interpret the model's behavior and identify physiologically relevant patterns, we employ two explainability strategies:

Gradient-weighted Class Activation Maps (Grad-CAM): To interpret which parts of the physiological signals contribute most to the model's decision, we apply Grad-CAM to the final convolutional layers of the architecture. This method, originally proposed by Selvaraju et al. [37], produces time-resolved activation maps that highlight temporal regions most influential for the predicted class. Technically, Grad-CAM computes the gradients of the score for the predicted class y^c (prior to softmax or sigmoid) with respect to the feature maps A^l of a given convolutional layer l. These gradients are globally averaged over the samples ($n = 1, ..., N$) to produce the importance weights α_l^c:

$$\alpha_l^c = \frac{1}{N} \sum_{n=1}^{N} \frac{\partial y^c}{\partial A_n^l}$$

The weighted combination of the feature maps is then passed through a ReLU activation to obtain the final class-discriminative localization map $\mathcal{M}^c_{\text{Grad-CAM}}$:

$$\mathcal{M}^c_{\text{Grad-CAM}}[n] = \text{ReLU}\left(\sum_l \alpha_l^c A_n^l\right)$$

This results in a temporal heatmap indicating which samples are most strongly influenced the model's output for the class c. The ReLU ensures that only the positive influences on the class score are retained. To enhance interpretability, the resulting Grad-CAM maps are upsampled to the original input resolution (1000 samples) and overlaid on the raw physiological signals.

Feature ablation: We systematically evaluate the individual contribution of each physiological modality through a channel-wise ablation procedure. Feature ablation is a standard technique in explainable AI to assess input importance by perturbation [18, 36]. Basically, for each validation sample, we perform three forward passes through the trained network: one with the complete input (PPG, EDA, and RESP), and three additional passes where each channel is independently set to zero while keeping the other two channels intact. The impact of each modality is quantified by measuring the change in the output probability distribution when that specific channel is ablated. This approach allows us to directly assess the relative importance of each physiological signal for pain classification decisions and identify which modalities are most critical for discriminating between different pain intensities.

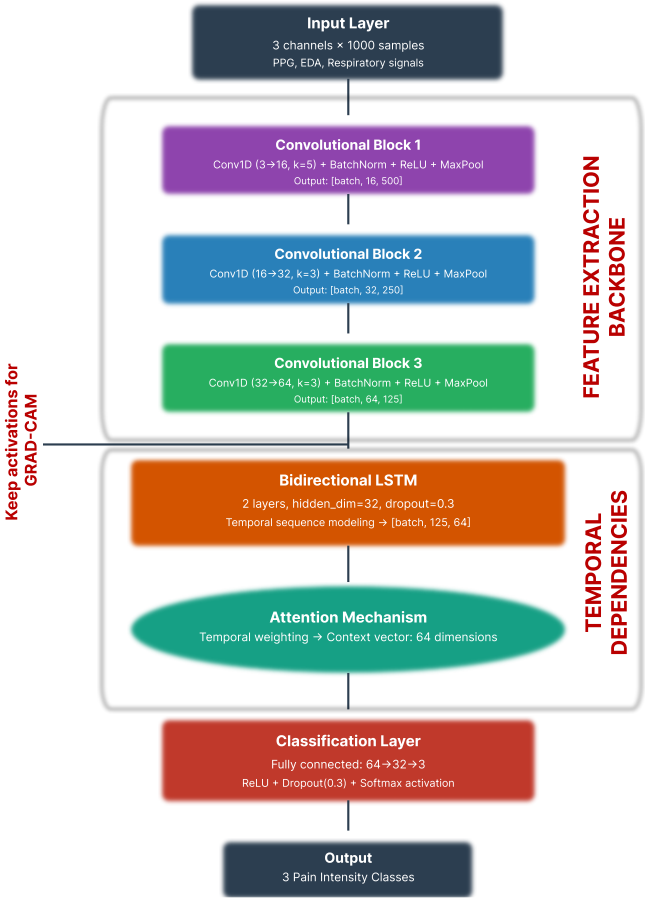


Figure 1: CNN-LSTM Pain Classification Architecture

2.2.2 Stage 2: XAI-Guided Feature Engineering. After carefully reviewing the Stage 1, we propose a comprehensive set of physiologically-informed features that uses the temporal and modal patterns revealed by Grad-CAM and feature ablation analysis. We extract features for each 10s segment as a whole. The feature engineering process is guided by two key findings: (1) PPG signals provide stable baseline information across all pain intensities, (2) EDA signals

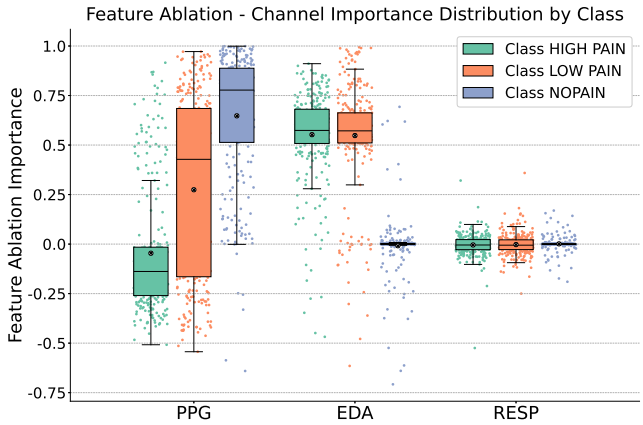


Figure 2: Channel importance distribution by pain class through feature ablation analysis. Box plots show the probability degradation when each physiological signal is ablated (higher values indicate greater importance).

exhibit progressive importance scaling with pain severity. The specific pipeline for obtaining the biomarkers are explained below in Sec. 2.3.

2.2.3 Stage 3: Building the ML Ensemble. With the XAI-guided feature set established, we proceed to construct a robust ensemble classifier through systematic hyperparameter optimization and model selection. The ensemble building process employs a three-phase approach: (1) comprehensive hyperparameter optimization across multiple algorithms, (2) patient-aware cross-validation for robust performance estimation, and (3) intelligent ensemble construction.

Hyperparameter Optimization Framework: We implement a Bayesian optimization strategy using Optuna's Tree-structured Parzen Estimator (TPE) sampler to efficiently explore the hyperparameter space [1]. The optimization targets twelve distinct algorithms including gradient boosting methods XGBoost [10], LightGBM [20], CatBoost [32] and ensemble method Random Forest [6]. For each algorithm, we define comprehensive search spaces tailored to their specific architectures: XGBoost and LightGBM optimization spans 50-500 estimators, learning rates from 0.01-0.3 (log-scaled), tree depths of 3-15, various subsampling ratios (0.6-1.0), and L1/L2 regularization parameters (0-10), with LightGBM additionally optimizing leaf count (10-300) and minimum child samples (5-100). CatBoost optimization includes iteration counts (50-500), tree depth (3-10), L2 leaf regularization (1-10), and algorithm-specific parameters such as border count (32-255) and bagging temperature (0-1). Random Forest hyperparameters encompass estimator count (50-500), maximum depth (3-20), minimum samples for splitting (2-20) and leaf nodes (1-10), feature selection strategies (sqrt, log2, all), and bootstrap sampling options. The optimization process employs early stopping with median pruning to accelerate convergence, executing 100 trials per model with a 30-minute timeout. Class imbalance is addressed through balanced weighting strategies specific to each algorithm.

Feature Space Optimization: Prior to model training, we apply a multi-stage feature selection pipeline targeting the engineered feature set from Stage 2. The initial feature space incorporates multiple normalization strategies to capture different aspects of physiological signal variability: raw unnormalized features, patient-wise normalized features using RobustScaler, MinMaxScaler, and StandardScaler (accounting for individual baseline differences), and globally normalized features using the previously mentioned scaling methods (capturing population-level distributions), resulting in approximately 400 total engineered characteristics across all normalization variants. The selection process then combines: (1) variance thresholding to remove low-variance features (threshold=0.01), (2) correlation filtering to eliminate redundant features (r>0.95), (3) univariate statistical testing selecting the top 70th percentile of features, and (4) model-based importance ranking using XGBoost with median threshold selection [17]. This comprehensive approach allows the algorithms to automatically determine which normalization strategy is most informative for each physiological feature, typically reducing the feature space to 100-150 highly discriminative characteristics while maintaining both physiological interpretability and normalization diversity. Given the previous criteria, we ultimately select the 100 most important features for our objective.

Ensemble Construction: Based on validation performance, we select the top-3 performing models to construct a heterogeneous ensemble. The ensemble employs probability averaging across diverse algorithms to maximize prediction robustness.

$$P_{ensemble}(y_i = c) = \frac{1}{3}\sum_{m=1}^{3} P_m(y_i = c), \tag{1}$$

where $P_m(y_i = c)$ represents the probability that observation i belongs to class c according to model $m = 1, 2, 3$.

After hyperparameter optimization, all selected models are retrained on the combined training and validation datasets to maximize available information for final predictions.

2.3 Physiological Feature Extraction

A comprehensive set of features was extracted from each physiological modality. The process for each signal is detailed below.

2.3.1 Preprocessing. Our pre-processing for PPG involved bandpass filtering using a 5th-order Chebyshev Type II filter (0.2-8 Hz) [27]. These filters were applied with a zero-phase approach in order to avoid signal distortion. The RESP is filtered by a linear detrend of the signal. A quality control step was implemented for flagging segments with EDA saturated more than 25% of the time or very poor PPG quality. During feature extraction, values from these flagged segments were imputed to prevent data contamination.

2.3.2 PPG Feature Extraction. The feature extraction pipeline for the PPG signal was designed to capture a wide range of cardiovascular dynamics. The process involved four main stages: signal pre-processing, pulse-level analysis, segment-level feature calculation, and signal quality assessment.

Following filtering and artifact rejection, each individual PPG pulse was automatically delineated to identify key fiducial points [25] for each pulse k: the maximum velocity instant in the systolic wave

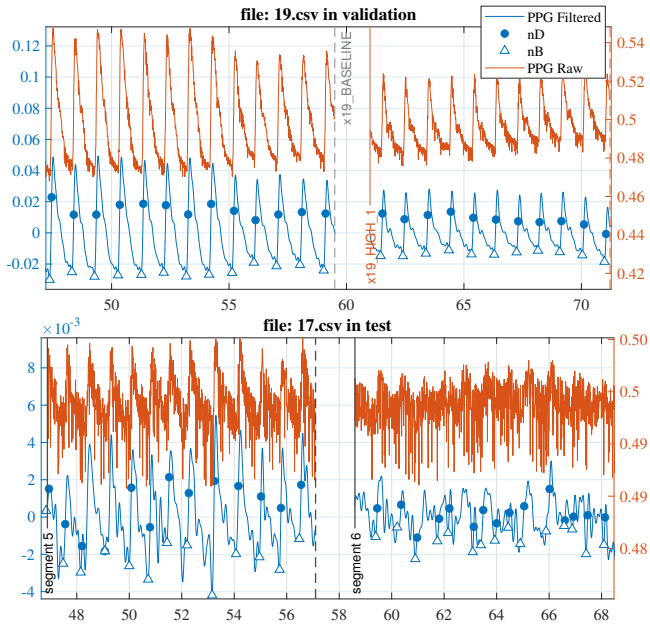


Figure 3: PPG signals with (a) good, and (b) bad signal quality.

($n_{\mathrm{U}}[k]$), the pulse onset ($n_{\mathrm{B}}[k]$), and the systolic peak ($n_{\mathrm{A}}[k]$). From these points, continuous time-series of pulse-to-pulse features were derived. These included:

- **Pulse Rate Variability (PRV):** The variation in the pulse-to-pulse interval: $d_{\mathrm{PRV}}(n) = \sum_{k=2}^{K}(n_{\mathrm{U}}[k] - n_{\mathrm{U}}[k-1])\delta(n - n_{\mathrm{U}}[k])$.
- **Pulse Amplitude Variability (PAV):** The pulse-to-pulse variation in amplitude between the systolic peak and the pulse onset: $d_{\mathrm{PAV}}(n) = \sum_{k=1}^{K}(x_{\mathrm{PPG}}(n_{\mathrm{A}}[k]) - x_{\mathrm{PPG}}(n_{\mathrm{B}}[k]))\delta(n - n_{\mathrm{B}}[k])$.
- **Pulse Decomposition Analysis (PDA):** Each cardiac cycle can be modeled as a sum of three constituent Gaussian waves [23], representing a primary wave and two reflected waves. This analysis yielded continuous time-series describing the amplitude, width, and relative timing of these underlying waves.

The core feature set was generated by partitioning the previously described morphological and variability time-domain series (i.e., PPG, PRV, PAV, PDA) into 10-second windows. Within each window, a rich set of statistics was computed, categorized as follows:

- **Time-Domain Statistics:** Basic statistical moments (mean, standard deviation, median, range), percentiles (10th, 25th, 75th, 90th), and inter-percentile ranges. The slope of a linear regression fit to the feature over the window was also computed to capture local trends. The Root Mean Square of Successive Differences (RMSSD) was also computed as a standard measure of short-term PRV.
- **Frequency-Domain Statistics:** The feature signal within each window was resampled to 4 Hz and analyzed using a Fast Fourier Transform (FFT). From the resulting power spectrum, we calculated the power in the low-frequency (LF: 0.04–0.15 Hz) and high-frequency (HF: 0.15–0.4 Hz) bands. The normalized LF power was computed as $P_{LFn} = P_{LF}/(P_{LF} + P_{HF})$.
- **Non-linear Features:** To quantify signal regularity, Sample Entropy and Fuzzy Entropy were calculated.

To address signal corruption, we computed Signal Quality Indices (SQIs) for each 10-second segment [12], including statistical (SNR, skewness, kurtosis), spectral (spectral entropy), and complexity (Hjorth parameters) metrics [8]. Additionally, a template-matching SQI was derived by correlating each pulse with both a global (cross-subject) and a subject-specific canonical pulse [26]. High correlation values indicated clean, stable morphology. All SQIs were included as features in the final model and were further analyzed between correctly and incorrectly labeled samples on the validation set to assess whether PPG signal quality affects prediction performance, using the Mann–Whitney U test.

2.3.3 EDA Feature Extraction. EDA signal was processed to extract features reflecting both tonic and phasic responses, which are closely linked to emotional arousal and pain perception.

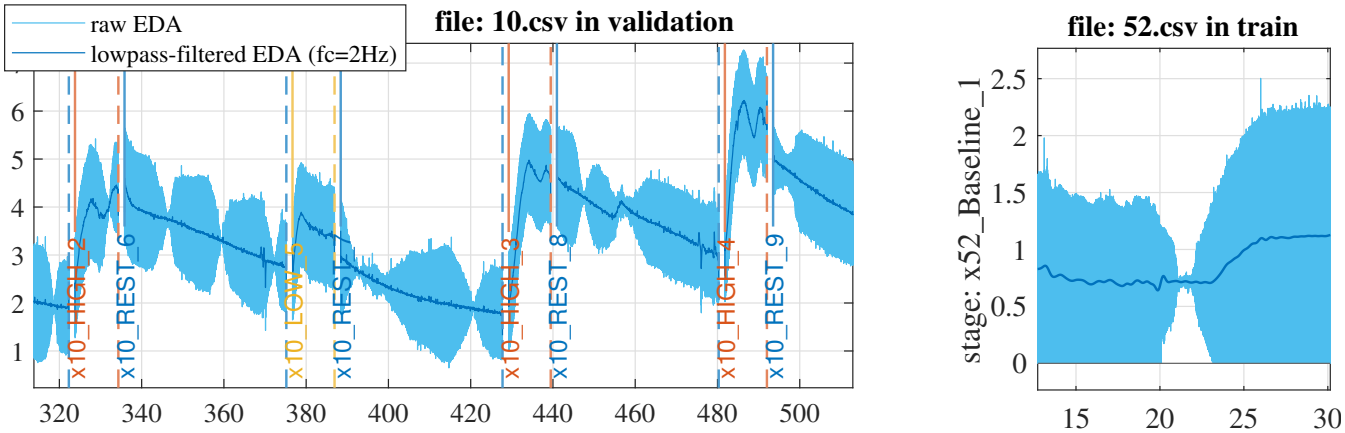
The cvxEDA[15] algorithm was used to decompose the raw EDA signal mainly into its tonic, sparse SMNA, and phasic components. This convex optimization-based method separates the slow-varying tonic component (Skin Conductance Level, SCL) from the faster phasic responses (Skin Conductance Responses, SCRs), which are assumed to be more directly related to transient stimuli such as pain. The following features were extracted:

- **Tonic Component (SCL):** Mean and slope of the tonic component, reflecting baseline sympathetic tone and slow temporal trends.
- **Phasic Component (SCRs):** Number of significant SCR peaks, mean amplitude, latency, and recovery time. These features capture the magnitude and frequency of rapid autonomic responses associated with acute pain.
- **Complexity Measures:** Sample Entropy and spectral entropy to quantify the irregularity and frequency content of the EDA signal.
- **Significative Sparse:** This metric counts the number of sparse activations in the driver signal p that exceed a threshold $p > 1$[15]. It serves as an indicator of the amount of meaningful sympathetic activations within the segment. A higher count reflects a more reactive phasic system, which is often associated with acute stress or pain stimuli. This feature is particularly useful for capturing transient autonomic responses to noxious events.

These features provide a detailed representation of autonomic activation patterns and are crucial for discriminating pain from non-painful states in downstream models.

2.3.4 RESP Feature Extraction. It implements metrics based on Hjorth parameters, peakedness, and other statistical descriptors to quantify signal variability and regularity. The primary features extracted include:

- **Amplitude metrics:** Extract information related to the amplitude of the signal and its derivative. These parameters try to find deep breaths that can be related to the recovery from pain and short breath as a reaction to something painful.



(a) EDA evolution during a good recording.

(b) EDA segment showing saturation.

Figure 4: Examples of EDA signals. (a) Evolution of EDA in a good recording. (b) A critical data quality issue where the signal flattens at the sensor's minimum measurable value (saturation). Such "artifacts" shown in (b) lead to sudden non-physiological increases in the filtered signal, that corrupt biomarkers' extraction and model predictions.

- **Mobility and Complexity (Hjorth parameters):** These quantify the frequency content and the rate of change in that frequency [8]. Increased mobility may reflect irregular or shallow breathing patterns, often observed during discomfort or stress.
- **Peakedness:** Measures the prominence of respiratory peaks within a segment [24]. This metric helps identify abrupt breathing patterns.

3 Results

The challenge committee evaluates performance primarily using the accuracy metric computed by the organizers on the withheld test set. For comprehensive analysis, we report precision, recall, F1-score, and accuracy across all three pain intensity levels (No-Pain, Low, High) on the validation set, while test set results are limited to accuracy scores provided by challenge organizers. In order to ensure fairness and consistency with the test data, we segment the validation signals into non-overlapping windows of 10 seconds each. Our three-stage methodology demonstrates progressive performance improvements, validating the effectiveness of XAI-guided feature engineering and ensemble learning approaches.

Feature ablation analysis reveals distinct physiological response patterns across pain intensities (Figure 2). PPG demonstrates consistently high importance, with a median of 0.75, establishing it as a stable baseline marker for pain detection. In contrast, EDA exhibits a systematic increase in importance from low to high pain, reflecting a possible progressive sympathetic nervous system activation correlated with pain intensity. This gradient pattern makes EDA the most discriminative modality for pain intensity classification. Respiratory signals did not show importance, median around 0 along all classes.

Figure 5 illustrates the explainability analysis of our multimodal pain classification model through Grad-CAM visualizations. The model demonstrates physiologically meaningful patterns that align with known pain responses. For *No Pain* samples (left figure), the model primarily attends to the pulse onsets in the PPG signal, which suggest it is focusing on HR metrics as an indicator of pain absence. Conversely, *High Pain* samples (right figure) show ongoing Grad-CAM activation, primarily focusing on PPG morphology at sections with elevated amplitude on the EDA channel. The model further detects irregularities in cardiac rhythm and changes in pulse amplitude variability. *Low Pain* samples (middle figure) exhibit patterns comparable to those of *High Pain* samples, although the EDA excitation appears to be of shorter duration for the lower pain instances. These patterns extend over the majority of the subjects.

Table I presents the performance progression across our three-stage methodology on the validation set. The baseline CNN-LSTM model (Stage 1) establishes a foundation with 0.65 accuracy (F1-macro: 0.66), demonstrating the effectiveness of attention-based temporal modeling for physiological signal classification, notably enhancing the basic baseline multimodal model accuracy of 0.55 provided by the organization. Stage 2 introduces XAI-guided feature engineering combined with a non-hyperparameter-optimized Random Forest classifier, yielding a substantial improvement to 0.80 accuracy (F1-macro: 0.79). This result highlights the superiority of traditional machine learning architectures over complex deep learning models when limited training data is available, as the simpler Random Forest model uses the engineered features to outperform the deep neural network by 13 points in terms of F1-macro score. The final ensemble approach (Stage 3) combines optimized gradient boosting algorithms including LightGBM, CatBoost, and XGBoost, achieving the highest validation accuracy of 0.84 (F1-macro: 0.84). The ensemble demonstrates balanced performance across all pain intensity levels with F1-scores exceeding 0.80 for each class (No-Pain: 0.89, Low: 0.81, High: 0.83), validating the effectiveness of combining complementary ML algorithms for robust physiological signal classification.

Table 1: Performance Progression Across Three Stages on Validation Set

Models	Precision				Recall				F1-score			
	NP	Low	High	Avg	NP	Low	High	Avg	NP	Low	High	Avg
Baseline (Multimodal)	-	-	-	0.55	-	-	-	-	-	-	-	-
S1: CNN-LSTM	0.63	0.62	0.70	0.65	0.74	0.48	0.81	0.67	0.68	0.54	0.75	0.66
S2: Random Forest	0.92	0.73	0.75	0.80	0.82	0.74	0.78	0.78	0.87	0.74	0.77	0.79
S3: ML Ensemble	0.90	0.80	0.83	0.84	0.88	0.81	0.83	0.84	0.89	0.81	0.83	0.84

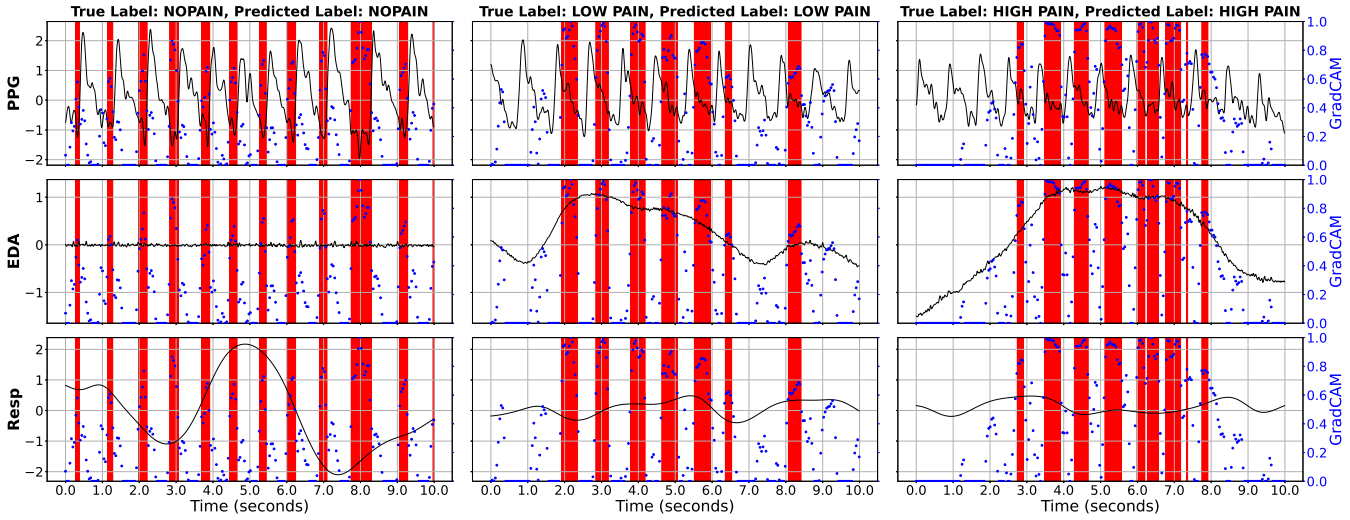


Figure 5: GradCAM visualization of the CNN-LSTM-Attention model's focus regions across three pain levels. Each row represents a different pain class (No Pain, Low Pain, High Pain) with synchronized PPG, EDA, and respiration signals over a 10-second window. Red overlays indicate regions of high model activation (GradCAM values above 70th percentile), while blue dots show the original 250 GradCAM values before temporal upsampling.

Moreover, to analyze the feature importance and validate the effectiveness of our XAI-guided feature engineering approach, we examine the feature importance distribution identified by the optimized XGBoost model. Table 2 presents the signal modality contribution ranking, revealing the dominant role of cardiovascular features and the relative contribution of different physiological modalities to pain classification performance. Furthermore, to evaluate our multi-normalization strategy's effectiveness, we conduct a similar importance analysis targeting normalization methods rather than signal modalities. The results demonstrate that patient-wise normalization dominates the feature landscape, accounting for 77% of the total discriminative power, while global normalization contributes only 12% and non-normalized features represent the remaining 11%.

Despite preliminary analysis suggesting the respiratory signal was less relevant, we included its features for two primary reasons. First, the short, segmented format of this dataset may mask the signal's true potential, which could be significant in applications with continuous monitoring. Second, our workflow was designed so that the data-driven feature selection process would objectively determine if these metrics provided useful information. Ultimately, this process confirmed that only a small subset of the respiratory features was informative for the final model.

Table 2: Feature Importance by Signal Modality

Signal	% Importance	Top 2 Features (% Importance)
PPG	80%	Pulse Amplitude Variability (10.9%) First Wave Amplitude 10th Percentile (7.6%)
EDA	15.5%	Signal Power (2%) Significative Sparse (1.6%)
RESP	4.5%	Mean Amplitude (1%) Peakedness of respiratory frequency (0.9%)

The test set performance of the Stage 3 ensemble model (0.70 accuracy with a 14% validation-to-test drop) reveals some generalization challenges, which represent a critical consideration for clinical deployment.

4 Discussion

Our results demonstrate the effectiveness of combining deep learning driven insight with classical feature-based machine learning for

pain recognition. The proposed three-stage approach successfully addressed the dataset's challenges, yielding significant performance gains over baseline models. Notably, the final ensemble achieved 84% accuracy on validation (F1-macro 0.84), markedly higher than the organizer's baseline (55% accuracy) and our initial CNN-LSTM model (65% accuracy). This highlights that, in data-constrained settings, carefully engineered physiological features and ensemble classifiers can outperform complex end-to-end deep networks. By leveraging deep learning in Stage 1 primarily for pattern discovery, we avoided the pitfalls of overfitting that often occur when training large models on limited data.

SQIs features indicate that PPG signal quality plays a role in prediction errors, although it is not the primary driver. The Mann-Whitney U test, applied to the validation set, revealed significant differences between accurate and inaccurate predictions for Zero Crossing Rate (p=0.015), Hjorth Complexity (p=0.020), and Dominant Frequency (p=0.028). This suggests a link between sensor errors and signal artifacts (see Fig. 3 and 4). We determined that these identifiable quality issues account for approximately 21% of the model's prediction errors. This indicates that enhanced quality-gating could offer a modest performance gain, and that there is a significant opportunity for improvement that lies in advancing the model's capacity to interpret the inherent physiological complexity present in biosignals of adequate quality. At the test stage, PPG's SQI decreased with respect to SQI at validation, being this one of the possible causes leading to increased prediction error.

The systematic performance progression across stages (0.65, 0.80, 0.84 validation accuracy) confirms that XAI-guided feature engineering provides more effective information transfer than end-to-end deep learning on limited datasets. By utilizing CNN-LSTM primarily for pattern discovery rather than direct classification, we avoided overfitting pitfalls common when training large models on small medical datasets [5]. The 15-point improvement from Stage 1 to Stage 2 demonstrates that carefully engineered physiological features can outperform or at least equal learned representations in data-scarce scenarios, consistent with recent findings in medical AI [35]. Our CNN-LSTM architecture with attention mechanisms proved effective for temporal physiological signal modeling [29], enabling meaningful pattern discovery through Grad-CAM visualizations [37]. Grad-CAM visualizations demonstrated that the CNN-LSTM model focuses on physiologically significant patterns: for pain, it focuses on PPG fiducial points while high amplitude in EDA is reached; whereas for NO-PAIN, the model targeted stable cardiac cycles and minor EDA amplitude variations.

Feature importance analysis concords with the previous XAI method, revealing patterns consistent with pain physiology literature while demonstrating our model's data-driven discovery capabilities. PPG signals dominate the feature landscape (79% total importance) through pulse amplitude variability and morphological features, reflecting cardiovascular responses to nociceptive stimuli [9]. While this contrasts with some studies emphasizing EDA dominance [31, 34], our findings suggest that cardiovascular dynamics provide the most discriminative information between No pain and the rest of the classes for this specific dataset and pain induction protocol. EDA contributes 15.5% of total importance through skin conductance responses and tonic components, confirming sympathetic nervous system activation during pain episodes. The systematic importance scaling observed in feature ablation validates the "fight-or-flight" response proportional to stimulus intensity. Respiratory signals provide an 8% contribution, consistent with their complementary role in autonomic pain responses [3]. The overwhelming preference for patient-wise normalization (77% vs 12% global importance) validates individualized baseline approaches critical for clinical deployment. This finding supports incorporating patient baseline calibration in pain assessment tools, as absolute physiological values vary dramatically between subjects due to individual pain tolerance and autonomic responsiveness differences; even gender can influence the physiological response to pain [40]. It should also be noted that, to reach its full potential, the cvxEDA library used for the analysis, as well as the extracted respiratory metrics, requires the signal to be recorded continuously for more than 10 seconds. This would enable the extraction of more robust metrics and allow for patient-wise calibration.

The interpretability framework addresses key requirements for the clinical adoption of AI [2]. Feature importance rankings provide transparent decision rationales, which are essential for building clinical trust [4]. This transparency enables healthcare providers to validate model predictions against established pain physiology, thereby supporting informed clinical decision-making—even in cases of variable model confidence. By grounding model behavior in genuine physiological phenomena rather than spurious patterns, the explainable framework is a critical step toward achieving trustworthy AI in medical applications

Despite strong validation performance, the 14-point test drop (0.84-0.70) highlights generalization challenges common in physiological machine learning [16]. Several factors may contribute to this performance gap that should be mitigated in future studies: (1) distribution shifts in test data with different artifact patterns not adequately represented in training, (2) inter-subject and intra-subject physiological variability exceeding training diversity, and (3) sensor quality issues including EDA saturation and PPG corruption. Also, the relatively small dataset size (65 subjects) limits model generalization capability, a common challenge in medical AI where labeled data is scarce [16].

5 Conclusion

In this paper, we presented a hybrid, three-stage methodology for objective pain assessment that bridges deep learning and classical machine learning. By using a DL model with XAI not as a predictor but as a discovery tool, we guided a targeted feature engineering process that fed a robust ensemble classifier. This approach, tailored to the challenges of segmented and noisy data, achieved state-of-the-art performance on the AI4PAIN 2025 Challenge dataset. The key takeaway is the value of using XAI as an integral component of the modeling pipeline, creating a pragmatic and powerful solution that capitalizes on the strengths of different modeling paradigms. This strategy offers a promising template for future work on challenging biomedical data.

Future steps could include a broader dataset with continuous signals and analyzing the scalability, stability, and generalization ability of the model.

6 Safe and Responsible Innovation Statement

This work addresses critical healthcare needs for objective pain assessment, particularly benefiting non-communicative patients. However, we acknowledge significant ethical considerations requiring careful attention. Our physiological monitoring approach raises privacy concerns regarding sensitive biometric data collection and storage, necessitating robust data protection protocols and informed consent procedures. The model's 14-point validation-to-test performance drop highlights generalization challenges that could disproportionately affect underrepresented populations, emphasizing the need for diverse training datasets and bias mitigation strategies. We strongly advocate that this technology should augment, not replace, clinical judgment, requiring extensive clinical validation before deployment. The explainable AI framework promotes transparency essential for medical applications, though potential misuse for surveillance or coercive pain assessment remains a concern. Future implementations must ensure equitable access across demographic groups, establish clear guidelines for clinical integration, and maintain healthcare provider oversight to prevent inappropriate automation of sensitive medical decisions.

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